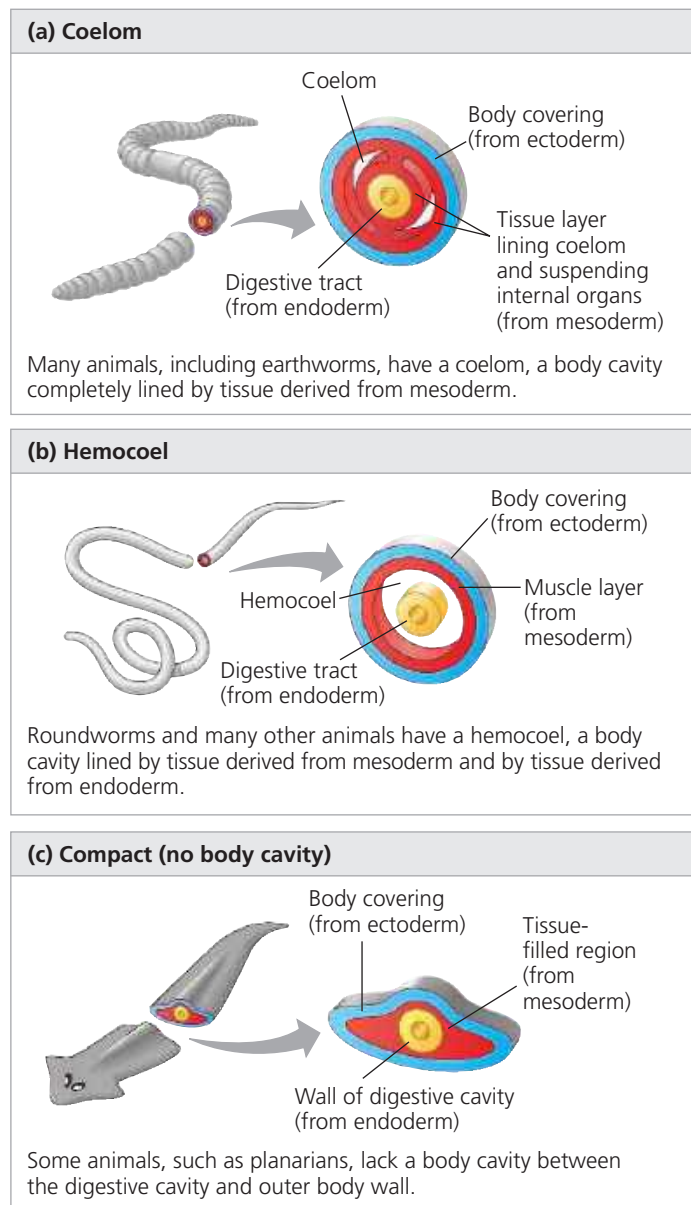


the internal organs (**Figure 32.9a**). A coelom's fluid cushions the suspended organs, helping to prevent internal injury. In soft-bodied animals, such as earthworms, the fluid in the coelom acts like a skeleton against which muscles can work. A coelom also enables the internal organs to grow and move independently of the outer body wall. If it were not for your coelom, for example, every beat of your heart or ripple of your intestine would warp your body's surface. Animals possessing coeloms are sometimes called *coelomates*.

Other triploblastic animals have a **hemocoel**, a body cavity that forms between the mesoderm and endoderm (**Figure 32.9b**). A hemocoel contains *hemolymph*, a fluid that functions in the internal transport of nutrients and waste.

▼ **Figure 32.9 Body cavities of triploblastic animals.** The organ systems develop from the three embryonic germ layers.



Key

■ Ectoderm ■ Mesoderm ■ Endoderm

Hemolymph is analogous to your blood and is circulated through the body cavity in an open system by the heart.

Many animals have a hemocoel and a coelom. Molluscs, for instance, have a hemocoel as the primary body cavity, but they also have a reduced coelom that surrounds the heart and reproductive structures (see Figure 33.16). Other animals, such as rotifers and nematodes (see Figures 33.12 and 33.26), have only a hemocoel. Animals with only a hemocoel once were called *pseudocoelomates* (from the Greek *pseudo*, false) and the cavity, a pseudocoelom. Hemocoels and coeloms have been independently gained or lost multiple times in the course of animal evolution; hence, their presence or absence is not a good indicator of phylogenetic relationships.

Finally, some triploblastic animals lack a body cavity altogether (**Figure 32.9c**). These compact animals tend to have thin, flat bodies. Such animals don't require an internal transport system: With bodies that are only a few cells thick, the exchange of nutrients, gases, and wastes can occur across the entire body surface. These animals are sometimes called *acoelomates* (from the Greek *a-*, without).

Protostome and Deuterostome Development

Many animals can be described as having one of two developmental modes: **protostome development** or **deuterostome development**. These modes can generally be distinguished by differences in cleavage, coelom formation, and fate of the blastopore.

Cleavage

Many animals with protostome development undergo **spiral cleavage**, in which the planes of cell division are diagonal to the vertical axis of the embryo; as seen in the eight-cell stage of the embryo, smaller cells are centered over the grooves between larger, underlying cells (**Figure 32.10a**, left). Furthermore, the so-called **determinate cleavage** of some animals with protostome development rigidly casts ("determines") the developmental fate of each embryonic cell very early. A cell isolated from a snail at the four-cell stage, for example, cannot develop into a whole animal. Instead, after repeated divisions, such a cell will form an inviable embryo that lacks many parts.

In contrast to the spiral cleavage pattern, deuterostome development is predominantly characterized by **radial cleavage**. The cleavage planes are either parallel or perpendicular to the vertical axis of the embryo; as seen at the eight-cell stage, the tiers of cells are aligned, one directly above the other (see Figure 32.10a, right). Most animals with deuterostome development also have **indeterminate cleavage**, meaning that each cell produced by early cleavage divisions retains the capacity to develop into a complete embryo. For example, if the cells of a sea urchin embryo are separated at the four-cell stage, each can form a complete larva. Similarly, it is the indeterminate cleavage of the human zygote that makes identical twins possible.